number_theory — User Guide

`number_theory` is a small, dependency-free tool for **GCD** and **LCM** (and the extended Euclidean algorithm). The math ships as a **compiled binary**, so you install and use it without any readable source for the algorithms.

You can use it two ways:

- as a **command-line tool** (`number-theory …`), or
- as a **Python library** (`import number_theory`).

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1. Requirements

The prebuilt release wheel targets:

Operating system	Windows, 64-bit
Python	CPython **3.14**, 64-bit

On any other OS or Python version, install by **building from source** (see 1.3) — you'll need a C compiler.

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2. Installation

> Tip: installing into a **virtual environment** keeps your global Python clean: `powershell python -m venv .venv .\.venv\Scripts\Activate.ps1 # Windows PowerShell`

2.1 From the GitHub Release (recommended)

```
pip install https://github.com/Kuantor/number_theory/releases/download/v0.1.0/number_theory-0.1.0-cp314-cp314-win_amd64.whl
```

Or download the `.whl` from the releases page and install the file:

```
pip install .\number_theory-0.1.0-cp314-cp314-win_amd64.whl
```

2.2 Verify the install

```
number-theory --version        # -> number_theory 0.1.0
python -c "import number_theory as nt; print(nt.gcd(12, 18))"   # -> 6
```

2.3 Build from source

Needs a C compiler (MSVC Build Tools on Windows; gcc/clang on Linux/macOS).

```
git clone https://github.com/Kuantor/number_theory
cd number_theory
pip install build
python -m build --wheel
pip install (Get-ChildItem dist\*.whl)    # PowerShell
# on macOS/Linux:  pip install dist/*.whl
```

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3. Command-line usage

```
number-theory <command> [numbers...]
```

Command	Meaning	Example	Output
`gcd N [N ...]`	Greatest common divisor	`number-theory gcd 12 18 24`	`6`
`lcm N [N ...]`	Least common multiple	`number-theory lcm 4 6 8`	`24`
`egcd A B`	Extended gcd: prints `g x y` with `A*x + B*y = g`	`number-theory egcd 240 46`	`2 -9 47`
`--version`	Print the version	`number-theory --version`	`number_theory 0.1.0`
`-h` / `--help`	Show help (works per command too)	`number-theory gcd -h`	usage text

Notes:

- `gcd` and `lcm` accept **two or more** integers.
- Negative integers are accepted; results are non-negative.
- If you run `number-theory` with no command it prints a usage error.

You can also run it without the console script:

```
python -m number_theory gcd 1071 462     # -> 21
```

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4. Python API

```
import number_theory as nt
```

Function	Returns	Description
`gcd(a, b)`	`int`	Greatest common divisor. Non-negative. `gcd(0, 0) == 0`.
`lcm(a, b)`	`int`	Least common multiple. Non-negative. `lcm(x, 0) == 0`.
`extended_gcd(a, b)`	`(g, x, y)`	`g == gcd(a, b)` and `a*x + b*y == g`.
`gcd_many(*nums)`	`int`	GCD of two or more integers.
`lcm_many(*nums)`	`int`	LCM of two or more integers.
`nt.__version__`	`str`	Installed version, e.g. `"0.1.0"`.

Examples

```
nt.gcd(12, 18)             # 6
nt.gcd(-12, 18)            # 6      (sign-insensitive)
nt.lcm(4, 6)              # 12
nt.lcm(0, 5)              # 0

nt.gcd_many(12, 18, 24)   # 6
nt.lcm_many(4, 6, 8)      # 24

g, x, y = nt.extended_gcd(240, 46)
# g == 2, and 240*x + 46*y == 2

# Arbitrary precision — works for huge numbers:
nt.gcd(2**512 * 3, 2**300 * 9)   # exact, no overflow
```

Input rules

- Inputs must be Python `int`. Passing a `float`, `str`, `None`, or `bool` raises `TypeError`.
- `gcd_many()` / `lcm_many()` with **no** arguments raise `ValueError`.

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5. Troubleshooting

`ModuleNotFoundError: No module named 'number_theory._core'` You're almost certainly running Python from **inside a clone of the repo**, so Python imports the local source folder (which has no compiled binary) instead of the installed package. Fix: run from a different directory, e.g.

```
cd ~
python -c "import number_theory as nt; print(nt.gcd(12, 18))"
```

`'number-theory' is not recognized as an internal or external command` The install worked — `pip` just placed the `number-theory.exe` launcher in your Python's `Scripts` folder, which isn't on your `PATH`. Three ways to deal with it:

Option A — run the module form (no PATH change needed):

```
python -m number_theory gcd 12 18
```

Option B — call the launcher by its full path. Find the folder with:

```
python -c "import sysconfig; print(sysconfig.get_path('scripts'))"
```

then run e.g. `& "<that folder>\number-theory.exe" --version`.

Option C — add that folder to your user `PATH` permanently (PowerShell):

```
$scripts = python -c "import sysconfig; print(sysconfig.get_path('scripts'))"
$u = [Environment]::GetEnvironmentVariable("Path", "User")
if ($u -notlike "*$scripts*") {
    [Environment]::SetEnvironmentVariable("Path", "$u;$scripts", "User")
    "Added - open a NEW terminal for it to take effect."
} else { "Already on PATH." }
```

Then open a **new** terminal and `number-theory --version` will work.

> Avoid the naive `setx PATH "%PATH%;..."` in `cmd`: `setx` truncates `PATH` at 1024 characters and folds your combined system+user `PATH` into the user variable, which can corrupt it. The PowerShell snippet above edits only the user scope and is safe.

If you're working inside a **virtual environment**, activating it (`.\.venv\Scripts\Activate.ps1`) already puts the launcher on `PATH` — no manual step needed.

`pip` says the wheel is "not a supported wheel on this platform" The release wheel is for Windows x64 + Python 3.14 only. On a different platform or Python version, build from source (see 2.3).

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6. Uninstall

```
pip uninstall number_theory
```

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7. Updates & links

- Releases: https://github.com/Kuantor/number_theory/releases
- Source & issues: https://github.com/Kuantor/number_theory
- Roadmap (upcoming: primality, factorization, totient, modular arithmetic): https://github.com/orgs/Kuantor/projects/10

This guide covers **v0.1.0**, whose scope is GCD / LCM only.